

Software Architecture and Design Thinking

116U01C701

Module 4

Module 4 (10)

4. Introduction to Design Thinking:

4.1 The Power of Design Thinking

Stages of thinking: The design process: Define, Research, Ideate, Prototype, Select, Implement, Learn, Example project

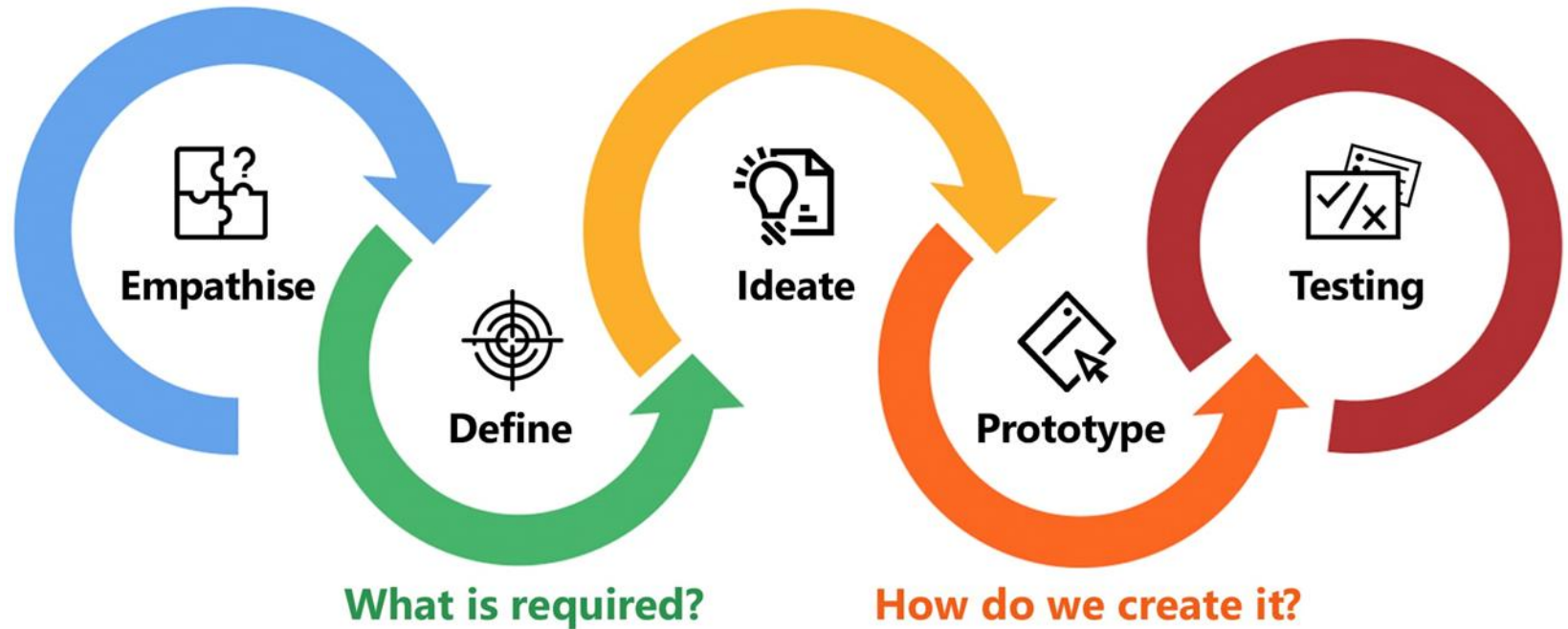
4.2 Research: Identifying drivers, Information gathering, Target groups, Samples and feedback

4.3 Idea generation: Basic design directions, Themes of thinking, Inspiration and references, Brainstorming, Value, Inclusion, Sketching, Presenting ideas

What is the Problem?

Why is it Important?

Does it work?



Introduction to Design Thinking

Design is an iterative process and design thinking is present in each stage of the journey from client brief to finished work.

Different solutions can be produced for any given brief and these can differ widely in levels of creativity, practicality and budget.

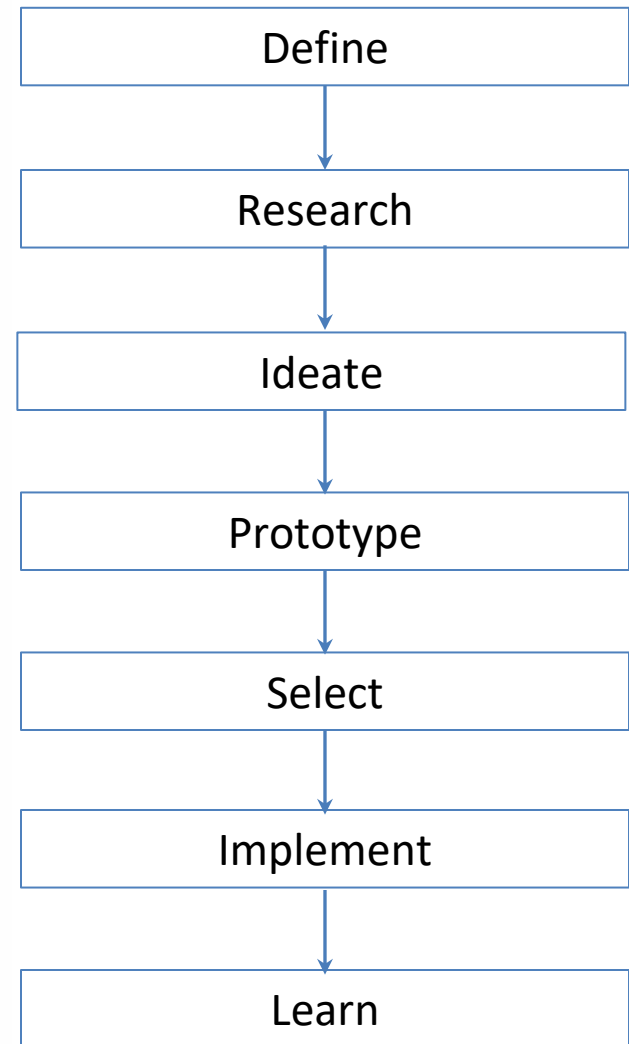
Stages of thinking

Design is a process that turns a brief or requirement into a finished product or design solution.

The design process can be said to comprise seven stages:

1. **Define**
2. **Research**
3. **Ideate**
4. **Prototype**
5. **Select**
6. **Implement**
7. **Learn.**

Each of these requires design thinking.



Stages of Thinking

The design process engages a high degree of creativity but in a way that is controlled and directed by the process so that it is channeled towards producing a viable, practical solution to the design problem, meeting or excelling the stated aims of the brief.

While creativity in design is important, design is an activity that serves economic as well as creative goals.

The design process helps ensure that a design satisfies all such considerations.

The process seeks to generate a number of possible solutions and utilizes various techniques or mechanisms that encourage participants to think outside the box in the pursuit of creative or innovative solutions



The design process

1. Define:

- The design problem and the target audience needs to be defined.
- A precise understanding of the problem and its constraints allows more exact solutions to be developed.
- This stage determines what is necessary for the project to be successful. The research stage reviews information such as the history of the design problem, end-user research and opinion-led interviews, and identifies potential obstacles

Define

1. Objectives:

- What the client hopes to achieve through commissioning a design job,
- It is important that these are fully understood and 'mapped' to your design thinking.
- Need to be specified so that the design team knows what it has to achieve.
- Get to the heart of the matter and focuses on
 - » what the client expects
 - » what the project boundaries are
 - » what deliverables are required

Design Process

1. Define

2. Research

- Collecting background information
- Can be either
 - **Quantitative:** with hard statistical numbers about the size and composition of target user groups
 - **Qualitative:** with information about what that user group buys or consumes and what their lifestyle is like.

It may be related to build a mental model of a typical user in order to enable the design team to obtain a good feel for what would appeal to them.

This includes factors such as education, career, holiday destinations, musical

Research

Primary research:

- The feedback generated during the learning phase of projects previously undertaken with the same or similar clients provides a starting point with regard to what worked and what did not work with a specific target group.

Secondary research

- The information obtained from general secondary sources such as consumer market research reports which provide the demographic breakdown and historic performance of given markets and market segments, and provide a clear view of how a market is structured.

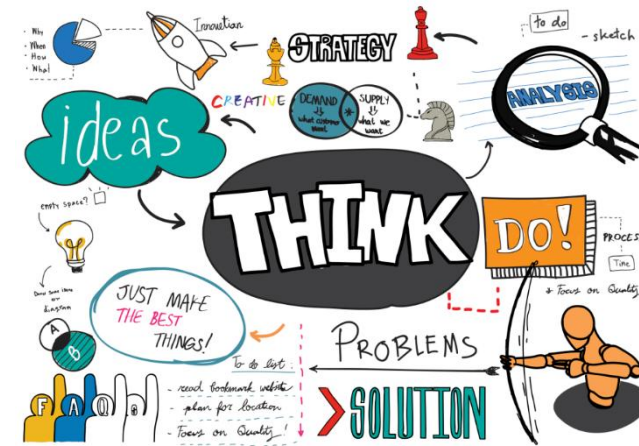
Design Process

1. Define
2. Research
3. **Ideate**

The design team draws on the research gathered and the constraints established during the define stage.

Ideation methods include

- brainstorming
- sketching ideas
- adapting a tried-and-tested design that already exists
- taking a top-down analytical approach that focuses on the product, service or company
- a bottom-up approach that focuses on the customer or user



Design Process

1. Define
2. Research
3. Ideate

4. Prototype

A prototype can:

- Be used to test the technical feasibility of a design idea to see if it works as a physical object
- Be used to test the visual aspects of the design by presenting them as they would be produced.
- Help to test particular aspects of a design solution, it must be made so that those aspects are present and can be effectively evaluated.
- Help in look and feel

Design Process

1. Define
2. Research
3. Ideate
4. Prototype

5. Select

The point at which one of the proposed design solutions is chosen for development that typically most closely meets the design brief, or a significant part of it.

Design Process

1. Define
2. Research
3. Ideate
4. Prototype
5. Select

6. Implement

The designer passes the design artwork and format specifications to those who will be supplying the final product

Ensure that the end results meet design expectations, within budget and time

Design Process

1. Define
2. Research
3. Ideate
4. Prototype
5. Select
6. Implement

7. Learn

Learning from what has happened throughout the design process

Becomes a learning opportunity for future projects

Although the learn stage appears to be the last of the seven that we've identified, it actually occurs throughout the design process

At each stage stock is taken of from where started and where heading, what's working and what's not.

The ability to learn from each stage will enhance the development of design thinking, and will help to generate radical and successful designs

Design Process

Example :

Project: Develop an interface for College Website

- 1. Define**
- 2. Research**
- 3. Ideate**
- 4. Prototype**
- 5. Select**
- 6. Implement**
- 7. Learn.**

Design Process

Example :

Project: Develop an interface for College Website

1. Define: The information to be displayed on the website

Identify personas:

- Probable students
- Parent of probable students
- Existing students
- Parents of existing students
- General Visitors
- Faculty
- Staff
- Recruiters

Check what they could be looking for

Design Process

Example :

Project: Develop an interface for College Website

2. Research

- Find how the existing websites look like
- Check websites of other universities/ colleges
- Understand the limitations of existing designs

Design Process

Example :

Project: Develop an interface for College Website

3. Ideate

- Prepare three – four designs considering the availability of technology, resources, time in hand and scope

Design Process

Example :

Project: Develop an interface for College Website

4. Prototype

Develop prototypes to get

- Look and Feel of the product
- Early inputs from the end user

Design Process

Example :

Project: Develop an interface for College Website

5. Select

Compare the various designs and choose the one which fits into the bill

The main deciding factor will be

- Meeting objectives **AND**
- User Experience

Design Process

Example :

Project: Develop an interface for College Website

5. Implement

Implement the selected model

Take inputs and approvals from end user from time to time

Design Process

Example :

Project: Develop an interface for College Website

6. Learn

Based on the inputs received during/ after implementation make changes in the design or note the findings from the current design for next project

4.2 Research:



- Identifying Drivers
- Gathering Information
- Defining Target Groups
- Sampling Leveraging Feedback

4.2 Research:

- Moves beyond assumptions to evidence-based insights.
- Informs every stage, especially "Empathize" and "Test."
- Reduces risk and increases the likelihood of user acceptance.
- Foundation for innovation that truly meets needs.



Identifying Drivers

- The "Why" Behind the "What"
- Underlying motivations, needs, and influences that shape user behavior.
- The psychological, personal, social, and cultural forces at play.
- Uncovering drivers helps explain *why* users do what they do, or *why* they need what they need.

Types of Drivers

- **Psychological Drivers:** Emotions, motivations, perceptions, attitudes, beliefs.
- **Personal Drivers:** Age, lifestyle, personality, economic status, occupation.
- **Social Drivers:** Family, friends, peer groups, social roles, cultural norms.
- **Cultural Drivers:** Values, customs, traditions, societal expectations.

Methods for Identifying Drivers

- **In-depth Interviews:** Asking "Why?" repeatedly (5 Whys).
- **Contextual Inquiry:** Observing users in their natural environment.
- **Empathy Maps:** Visualizing what users say, think, feel, and do.
- **Journey Mapping:** Tracing user experiences to find pain points and motivations.

Information Gathering & Synthesis

- Moving from data collection to insight generation.
- Articulating the core problem or opportunity in a human-centered way.
- **Output:** Point-of-View (POV) statements and "How Might We" questions.

Information Gathering & Synthesis

- Information Gathering for Building the Knowledge Base
- **Primary Research:** Data collected firsthand for the specific project.
 - Interviews, surveys, observations, focus groups.
- **Secondary Research:** Utilizing existing data.
 - Market reports, academic studies, competitor analysis, trend analysis.
- Combining both provides a holistic view.

Information Gathering & Synthesis

- Tools for Data Collection
- **Surveys:** Quantitative, broad reach, structured data.
- **Interviews:** Qualitative, deep insights, flexible.
- **Observation:** Uncovers unspoken needs, behavioral insights.
- **Focus Groups:** Group dynamics, diverse opinions.
- **Ethnography:** Immersive study of culture and behavior.

Target Groups

- Identifying the specific segment of users whose needs we are addressing.
- Moving beyond "everyone" to a clearly defined audience.
- Segmentation based on demographics, psychographics, geography, and behavior.
- **Why?** Tailor solutions, allocate resources effectively, communicate precisely.

Target Groups

- **Achieved through Creating User Personas**
 - Fictional, archetypal representations of ideal users.
 - Based on real data from research.
 - Includes demographics, behaviors, motivations, goals, and pain points.
 - Helps teams empathize and make user-centered decisions.

Samples

- Representing the Whole Users
- Selecting a subset of the target group for research.
- Ensuring the sample is representative to generalize findings.
- Crucial for both qualitative (deep dive) and quantitative (broad insights) research.

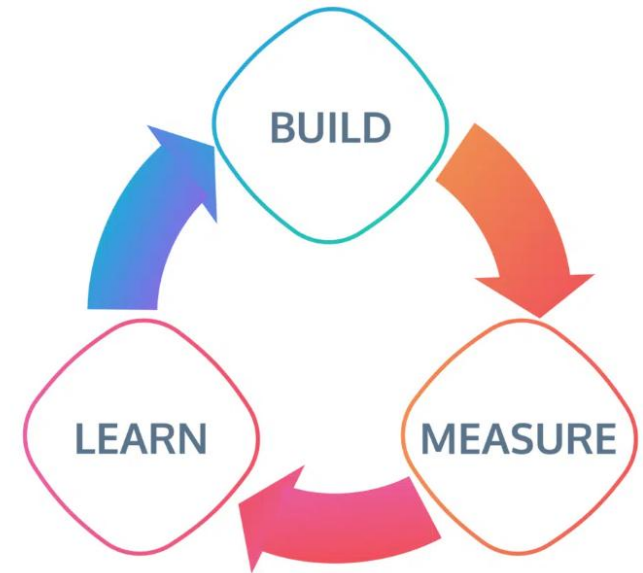
Sampling Methods

- Choosing Your Sample Wisely
- **Probability Sampling:** Random selection, higher generalizability (e.g., Simple Random, Stratified).
- **Non-Probability Sampling:** Non-random selection, often for exploratory research (e.g., Convenience, Purposive, Snowball).

Choice depends on research goals, resources, and desired rigor.

Feedback

- **Learning Through Iterations**
 - Presenting prototypes to real users to gather feedback.
 - Not about validating, but about learning and refining.
 - Crucial for uncovering usability issues and unmet needs.



Feedback

- **Helps in refinement**
 - Direct responses, opinions, and reactions from users.
 - Provides insights into what works, what doesn't, and what's missing.
 - Essential for iterating and improving design solutions.

Feedback

- Effective Feedback Loops
 - **Methods:** Usability testing, A/B testing, surveys, interviews (post-use), analytics.
 - **Analyze:** Look for patterns, common themes, unexpected behaviors.
 - **Synthesize:** Translate feedback into actionable insights.
 - **Iterate:** Use insights to refine prototypes and designs. This closes the loop back to the Empathize/Define phases.

4.3 Idea Generation:

- Basic design directions
- Themes of thinking
- Inspiration and references
- Brainstorming
- Value
- Inclusion
- Sketching
- Presenting ideas

Idea Generation

Once a design brief has been defined and researched, the idea generation or ideate stage sees concepts created that may solve the design problem

What is
**IDEA
GENERATION?**

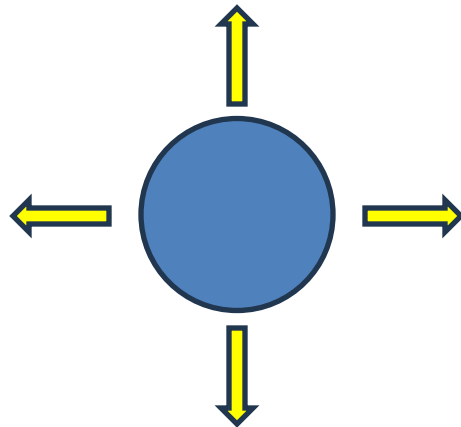


Idea Generation

— Basic design directions

- **Divergence:**

- The expansion or spreading out of something from a central point or theme.
- Divergence occurs as both an instigator and a response to divergence in society at large as designers respond to changing demographics, and the increasingly diverse market segmentations of their clients



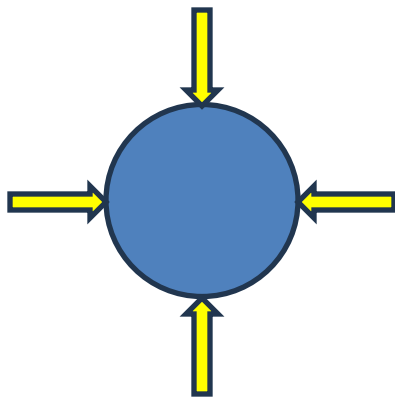
Moving away in different directions from a common point. Also called branching out.

Idea Generation

— Basic design directions

- **Convergence**

- The contraction of something towards a central, more generalized point
- Can still be found in areas such as generic branding.
- Brands nowadays often take products back to a more basic state or pre-branding time – when a tin of tomatoes was just a tin of tomatoes, for example.



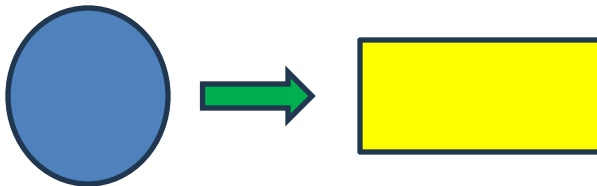
The coming together of two or more entities towards a central point or common ground.

Idea Generation

- **Basic design directions**

- **Transformation**

- Involves a substantial qualitative change, such as the redesign of a visual identity, in order to facilitate a new distribution method



A qualitative change in
appearance or character

Idea Generation

- **Basic design directions**
 - Designers compare existing design with competitors and then finalise about
 - Similar design (ease of use)
 - Contrasting design (to stand out)

Point of difference or unique selling point (USP)

The combination of values and attributes that differentiates a company or product from all other similar companies or products. As such, it is a form of divergence as it involves moving away from the designs used by similar companies or brands



Idea Generation

- Basic design directions
- **Themes of thinking**
- the challenge of fitting large quantities of information into formats with limited space
- **KISS** Keep It Short and Simple, or Keep It Simple Stupid (KISS)
- The idea is to trim a design to its essential elements, something that requires a clear understanding of the message that has to be communicated and the audience it is to be directed towards



Idea Generation

- Basic design directions
- **Themes of thinking**

- **Focus**

- Select only the key message elements as the focus for the design.
- design should focus on the most important ones which are relevant
- Other information may be shared on demand

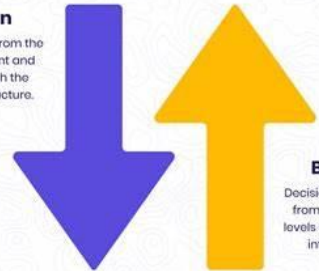
- **Top down and bottom up**

- An analytical approach appropriated from information technology development,
- this looks at a design problem from the system perspective and then 'drills' down
- to add detail in specific areas (top down), or focuses on the basic elements first
- and works upwards to link these together as part of a system (bottom up).



Top-down vs. bottom-up

Top-down
Decisions cascade from the upper management and filter down through the organizational structure.



Bottom-up
Decision-making originates from individuals at lower levels and gradually rises to influence the entire company.

 cascade

Idea Generation

- Basic design directions
- **Themes of thinking**
- **Ockham's razor Principle**
- states that elements that are not really needed should be pared back to produce something simpler
- The risk of introducing inconsistencies, ambiguities and redundancies will be reduced.
- Also referred to as the principle of parsimony or law of economy.

Occam's Razor

The main idea: Don't fabricate a complicated theory when you have a simpler one that handles a phenomenon just as well.

Idea Generation

- Basic design directions
- **Themes of thinking**
 - **Python philosophy**
 - beautiful is better than ugly;
 - simple is better than complex;
 - sparse is better than dense;
 - readability counts;
 - practicality beats purity; and
 - refuse the
 - temptation to guess

Idea Generation

- Basic design directions
- **Themes of thinking**
 - **White Space**
 - white space allows key design elements to breathe
 - Be easily seen
 - Helps the viewer to focus attention on them
 - Giving them greater impact

Idea Generation

- Basic design directions
- **Themes of thinking**
 - **Text minimization**
 - Text should be kept to a minimum
 - Sentences pared back to short
 - Sharp phrases that have a meaningful impact

Idea Generation

- Basic design directions
- **Themes of thinking**
 - **Graphic Impact**
 - Graphics should create a visual impact that grabs the attention and reinforces text communication.
 - Graphics that go overboard and are too large, complicated or numerous are distracting.
 - **Scale**
 - Design proofing needs to include an actual scale proof for small- or
 - large-scale items such as stamps or posters to ensure that text and graphics are of sufficient scale to be comfortably read.

- Basic design directions
- # Idea Generation

Themes of thinking

- ### User-centred design (UCD)

- User-centred design (UCD) places the needs, desires and limitations of the user at the centre of every stage of the design process
- requires designers to foresee how they are likely to use the resulting product.

- ### Ergonomics

- The practice of designing in accordance with physical human
- Optimise performance and minimise discomfort
- Focus on safety, efficiency, productivity and health in work settings to ensure that products, services and environments are compatible with the human form.

Idea Generation

- Basic design directions
- **Themes of thinking**
 - **TIMTOWTDI (pronounced Tim Toady)**
 - There Is More Than One Way To Do It'
 - A problem may have several different, but equally valid, solutions.

Idea Generation

Inspiration and references

- Inspiration is key to the generation of exciting design ideas
- people draw inspiration from both obvious and unexpected sources, such as magazines, music, literature and the urban environment
- The work of other people in the field, past and contemporary, provides creative stimulation

Idea Generation

- **Brainstorming Process:**
 - Define the problem to be addressed,
 - Select group participants that will address it
 - Form questions with which to stimulate the creative process.
 - Participants have free hand to make suggestions in a non-critical environment.
 - Encourage the presentation of unusual and potentially useful ideas.
 - Use resources such as flip charts or a whiteboard may be used to facilitate

Following the session,
ideas are grouped by type and their suitability assessed in order for a shortlist of the best ideas to be formed.

Idea Generation

- **Brainstorming Rules:**

- **Do not criticize:** Criticism prevents people from making suggestions and voicing options. Any idea is valid in brainstorming.
- **Keep the process manager-free:** the presence of line managers may inhibit the flow of ideas.
- **Avoid resolve:** do not start working up or resolving an idea that looks like a possible leader during the session. Instead, carry on generating ideas during the allotted time.
- Ideas can be resolved following the evaluation stage.
- **Work to a target:** a numerical target helps idea generation as participants move away from standard thinking on the subject in order to achieve it.
- **Focus on quantity not quality.**
- **Clock watch:** set a time by which the session should end. This helps to keep the pressure on, forcing more ideas to be generated.
- **Let go:** participants must not be afraid of offering odd, wacky or wild ideas, although this does not mean participants should not take the session seriously.
- **Be inclusive:** the session chairperson should prevent any group members from dominating the session and should encourage all members to contribute.

Idea Generation

- Basic design directions
- Themes of thinking
- Inspiration and references
- Brainstorming
- Value

Idea Generation

– Value

- A designer ‘adds value’ to a brand through the creation of a visual identity in terms of
 - **productivity**: the work completed in a given time
 - **boosting sales**
 - **increasing the prestige of a brand or organization**
- Testing with focus groups can give an indication of how positively the target group will respond to a design and its perceived value.
- Surveys can also provide feedback once a new design has been launched and will provide a learning opportunity about the value added by design.

Idea Generation

- **Inclusion**

- In order to judge how a target group will respond to a design, it is desirable to solicit the opinions of members of that group where possible and **include the target audience** in the ideate process.
- Solicitate the ideas, opinions and views of the target group.
- Give the target group an element of ownership of the ideas generated

This contribution helps ensure an idea resonates with the group, aiding acceptance, and reducing the possibility of errors by the design team, who may think they know what is best for the target group.

Idea Generation

- **Sketching**
 - convey a visual idea for a design or design element very rapidly
 - can form part of the definition stage as the brief may be sketch by the client
 - a designer rapidly outlines possible design solutions and creates a visual representation of ideas as they are generated

Idea Generation

- Presenting ideas
 - The candidate solutions is presented in such a way that the client can appreciate and understand the thought processes behind them and the messages they are trying to communicate

Idea Generation

- Basic design directions
- Themes of thinking
- Inspiration and references
- Brainstorming
- Value
- Inclusion
- Sketching
- Presenting ideas